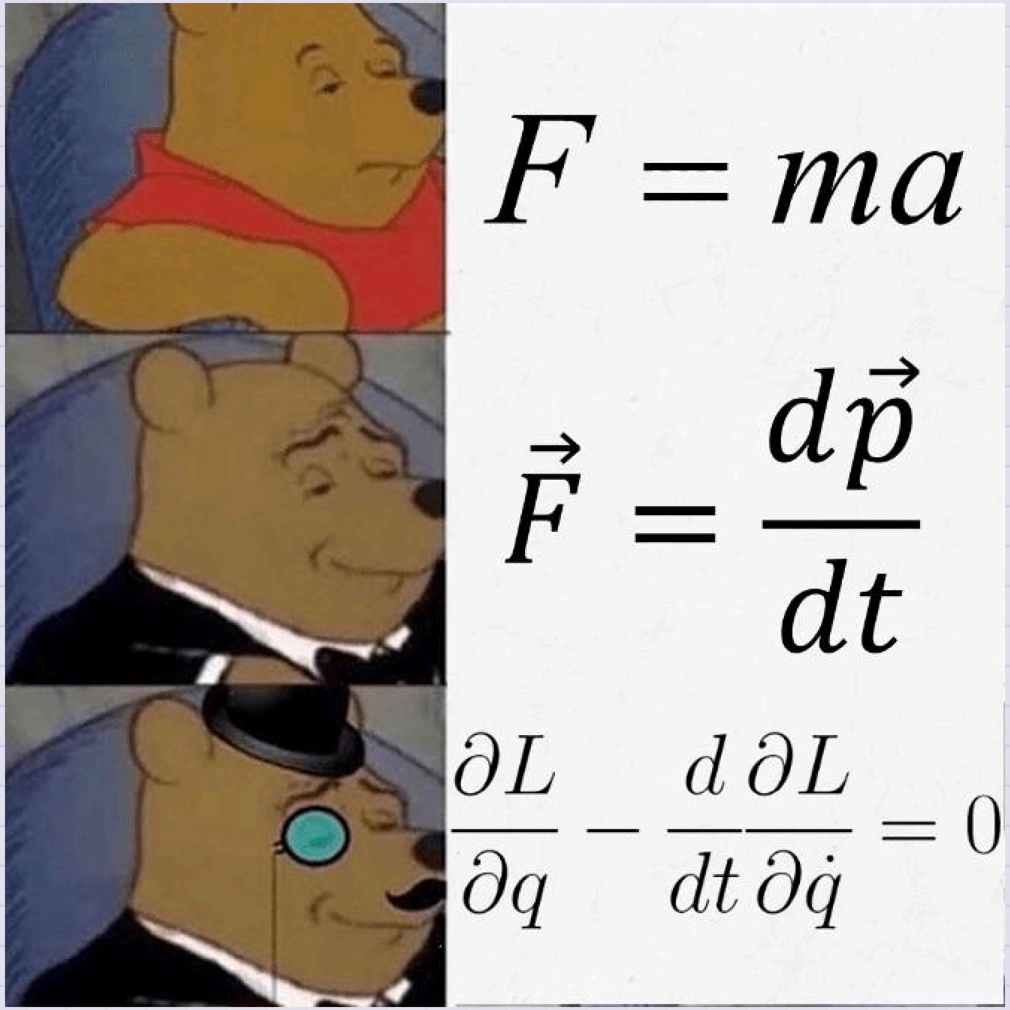
Day 35 - Lagrangian Examples



Announcements

- Last "Class" Week
- Homework 8 due Friday, Nov 21st (late after Nov 26th)
- Next Week: Project Work and Discussion
- Last Week: Presentations
- Final Project Due Dec 8th (no later than 11:59 pm)
- **No Final Exam**

Complete Google Form

By November 21st

Reporting your group members for the final project and a short summary of your project idea for sharing with the class.

https://forms.gle/iPKR9EDAaHW3GirN7

Reminders

We used the Lagrangian formalism to derive the equations of motion for a plane pendulum. We chose the x and y coordinates.

$$T(\dot{x},\dot{y}) = \frac{1}{2}m(\dot{x}^2+\dot{y}^2) \quad V(y) = mgy$$

$$\mathcal{L} = T - V = \frac{1}{2}m(\dot{x}^2+\dot{y}^2) - mgy$$

This gave us the following derivatives for the Lagrangian:

$$\frac{\partial \mathcal{L}}{\partial x} = 0 \quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) = \frac{d}{dt}(m\dot{x}) = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = -mg \quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{y}}\right) = -m\ddot{y}$$

We made a mistake by not including the constraint

We made a mistake by not including the constraint $x^2 + y^2 = L^2$ in our Lagrangian.

We can change variables to r and ϕ.

$$x = r\cos(\phi) \quad y = r\sin(\phi)$$

$$T(\dot{x}, \dot{y}) = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) = \frac{1}{2}m\left(r^2\dot{\phi}^2 + 2r\dot{r}\dot{\phi} + \dot{r}^2\right) = T(r, \dot{r}, \phi, \dot{\phi})$$

$$V(y) = mgy = mgr\sin(\phi) = V(r, \phi)$$

Now we include the constraint $r = L$, so that $\dot{r} = 0$.

$$T(\phi, \dot{\phi}) = \frac{1}{2}mL^2\dot{\phi}^2 \quad V(\phi) = mgL\cos(\phi)$$

$$\mathcal{L} = \frac{1}{2}mL^2\dot{\phi}^2 - mgL\cos(\phi)$$

Clicker Question 35-1

With $\mathcal{L} = \dfrac{1}{2} m L^2 \dot{\phi}^2 - mgL\cos(\phi)$, we can find the equations of motion.

$$\frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right) = 0$$

Which of the following equations of motion is correct?

1. $\ddot{\phi} = -\frac{g}{L}\sin(\phi)$
2. $\ddot{\phi} = -\frac{g}{L}\cos(\phi)$
3. $\ddot{\phi} = -\sqrt{\frac{g}{L}\sin(\phi)}$
4. $\ddot{\phi} = -\sqrt{\frac{g}{L}\cos(\phi)}$
5. None of these

Clicker Question 35-2

For the Atwood's machine, M is connected to m by a string of length l. Each mass has a length of string extended as measured from the center of the pulley (R) of y_1 and y_2, respectively. The string wraps around half the pulley.

Which of the following represents the equation of constraint for the system?

1. $y_1 + y_2 = l - R\phi$
2. $y_1 - y_2 = l + R\phi$
3. $y_1 + y_2 = l - \pi R$
4. $y_1 - y_2 = l + \pi R$
5. None of these

Take the time derivative of the constraint equation. What do you notice?

Clicker Question 35-3

With a Lagrangian of the form $\mathcal{L} = \frac{1}{2}(M+m)\dot{y}_1^2 - (M-m)gy_1$, we can find the **generalized forces** and **generalized momenta**.

$$F_{y_1} = \frac{\partial \mathcal{L}}{\partial y_1} = -\frac{\partial V}{\partial y_1} \quad p_{y_1} = \frac{\partial \mathcal{L}}{\partial \dot{y}_1} = \frac{\partial T}{\partial \dot{y}_1}$$

What are F_{y_1} and p_{y_1} for the Atwood's machine?

1. $F_{y_1} = -mg$ and $p_{y_1} = m\dot{y}_1$
2. $F_{y_1} = -Mgy_1$ and $p_{y_1} = M\dot{y}_1$
3. $F_{y_1} = -(M-m)g$ and $p_{y_1} = (M+m)\dot{y}_1$
4. $F_{y_1} = -(M+m)g$ and $p_{y_1} = (M-m)\dot{y}_1$
5. None of these

Clicker Question 35-4

Now, we allow the pulley (mass, M_p) to rotate. The Lagrangian is given by:

$$\mathcal{L} = \frac{1}{2}(M+m)\dot{y}_1^2 + \frac{1}{2}I\dot{\phi}^2 - (M-m)gy_1$$

Where I is the moment of inertia of the pulley. What is the moment of inertia of the pulley?

1. $I = \frac{1}{2}M_pR^2$
2. $I = \frac{1}{3}M_pR^2$
3. $I = M_pR^2$
4. $I = \frac{1}{4}M_pR^2$
5. None of these

Clicker Question 35-5

The rope moves without slipping on the pulley. A rotation of $Rd\phi$ corresponds to a displacement of dy_1 for the first mass, M. What is the **new** equation of constraint for the system?

1. $y_1 + y_2 = l - R\phi$
2. $dy_1 = Rd\phi$
3. $y_1 = R\phi$
4. $\dot{y}_1 = R\dot{\phi}$
5. More than one of these